Lecture 12(a): Need for updating DER meta data

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Agenda

- What is DER meta data?
- Use cases of DER meta data

What is DER Metadata?

- DER metadata: installed capacity, operating limits, geolocation, control & communication capabilities
- Aggregated DER metadata: total DER capacity downstream of LV substations

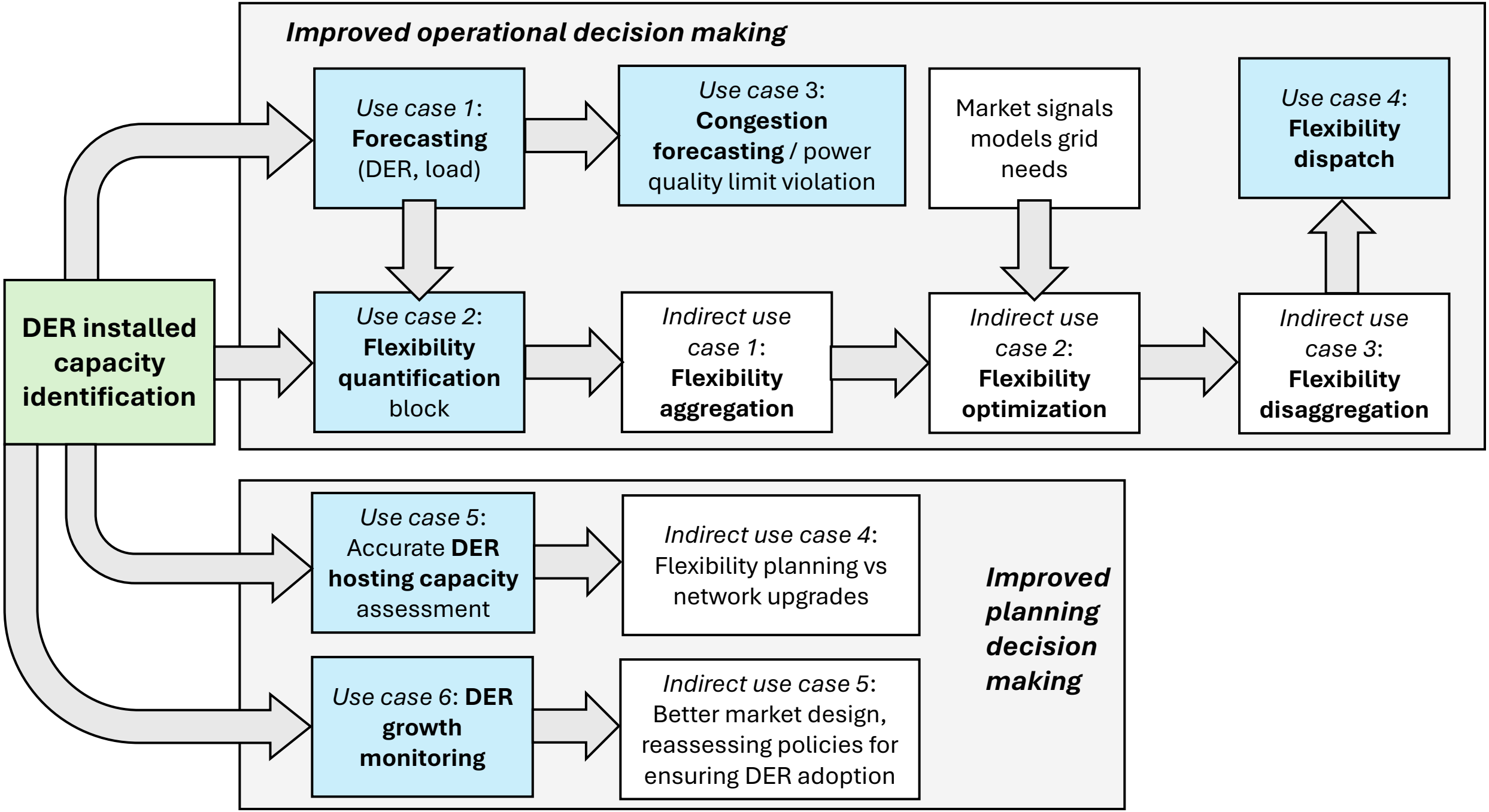
- Lecture focuses on installed capacity: PV (kWp), HP, EV chargers, BESS, etc.

Why Update DER Metadata?

- DER metadata degrades over time: upgrades, replacements, ownership changes
- DER penetration rapidly evolving → outdated registries mislead planning and operations
- Accurate metadata essential for forecasting, congestion management, flexibility, hosting capacity

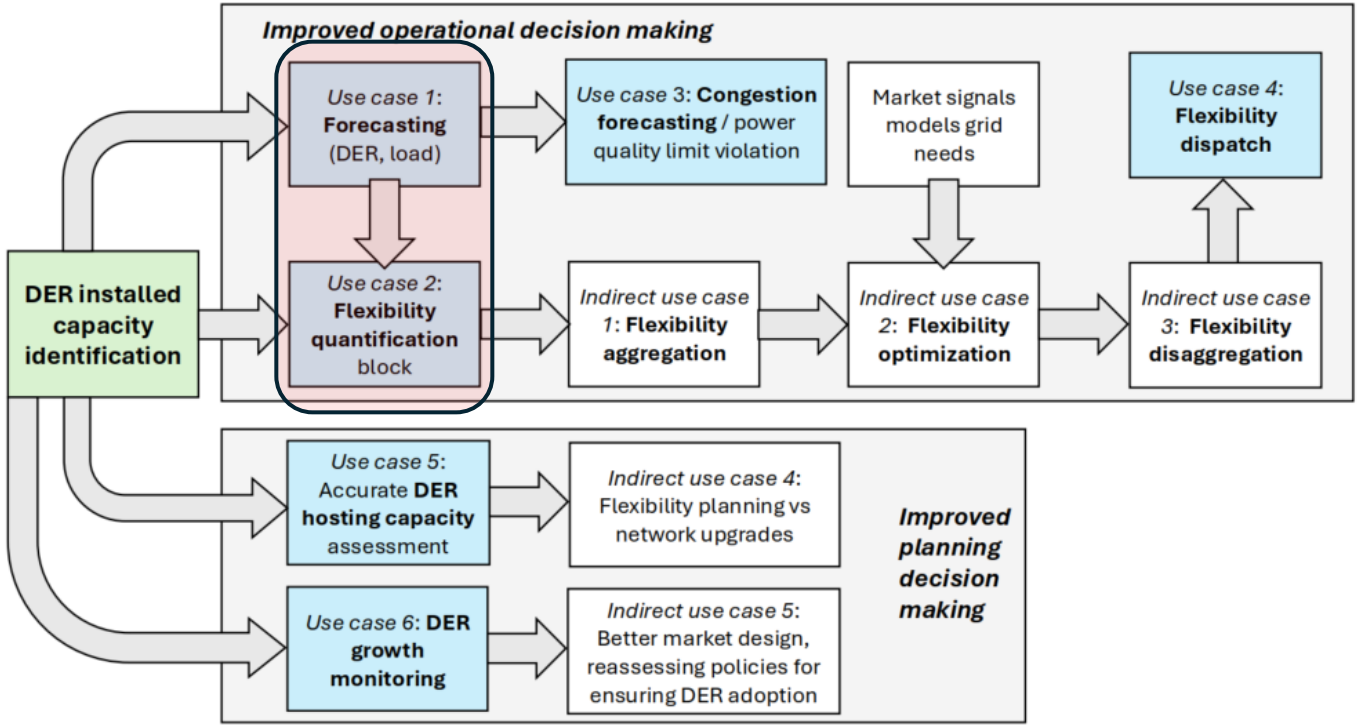
Why DSOs Lack Metadata?

- Incomplete observability: limited LV monitoring
- Missing/incorrect feeder topology, unknown phase allocation
- Regulatory/privacy constraints: restricted customer-level data
- DER installations not always reported (non-compliance)


Improved operational decision making
Use case 1: Forecasting (DER, load)
Use case 3: Congestion forecasting / power quality limit violation
Market signals models grid needs
Use case 4: Flexibility dispatch
DER installed capacity identification
Use case 2: Flexibility quantification block
Indirect use case 1: Flexibility aggregation
Indirect use case 2: Flexibility optimization
Indirect use case 3: Flexibility disaggregation
Use case 5: Accurate DER hosting capacity assessment
Indirect use case 4: Flexibility planning vs network upgrades
Improved planning decision making
Use case 6: DER growth monitoring
Indirect use case 5: Better market design, reassessing policies for ensuring DER adoption

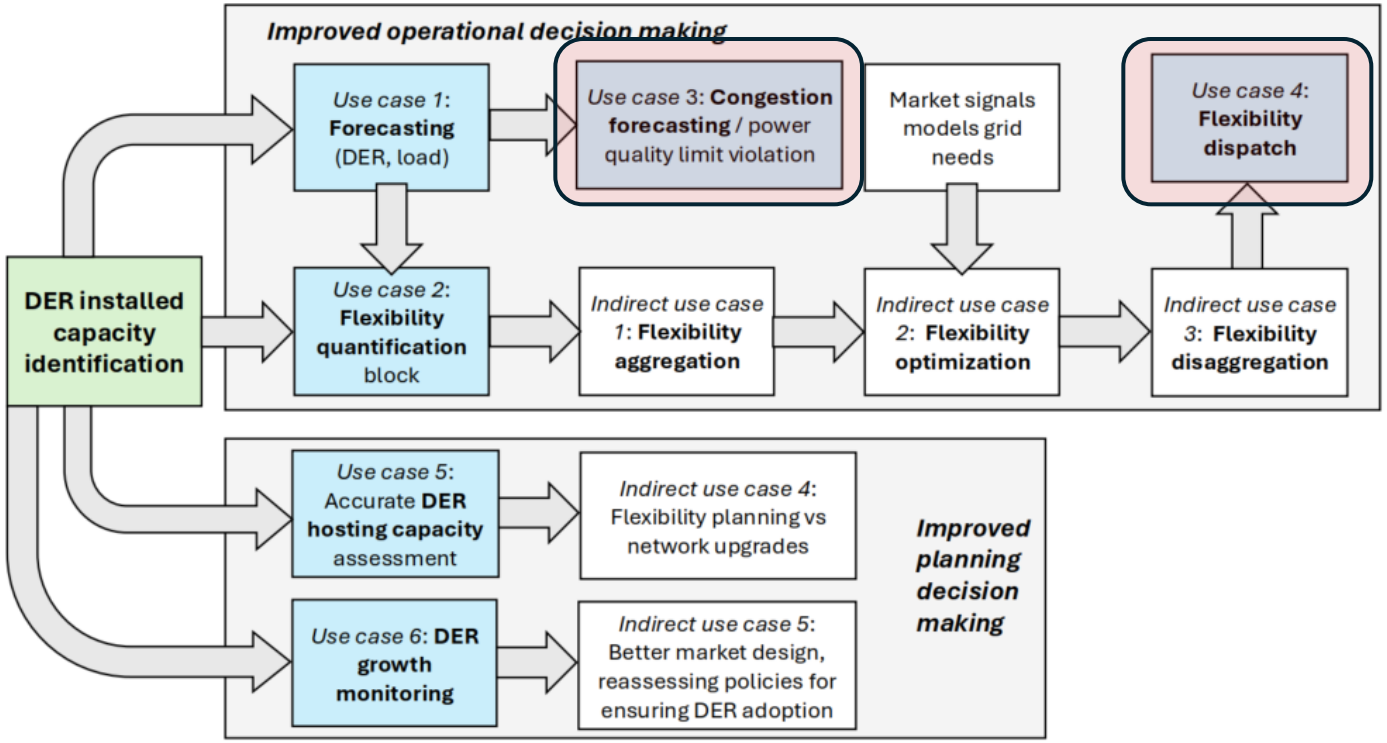
Use Cases 1–2: Forecasting & Congestion Management

- DER-aware forecasting using installed DER capacity improves accuracy by 11–16%
- Congestion prediction: DER metadata reveals synchronous DER behaviours→ peak loading risk



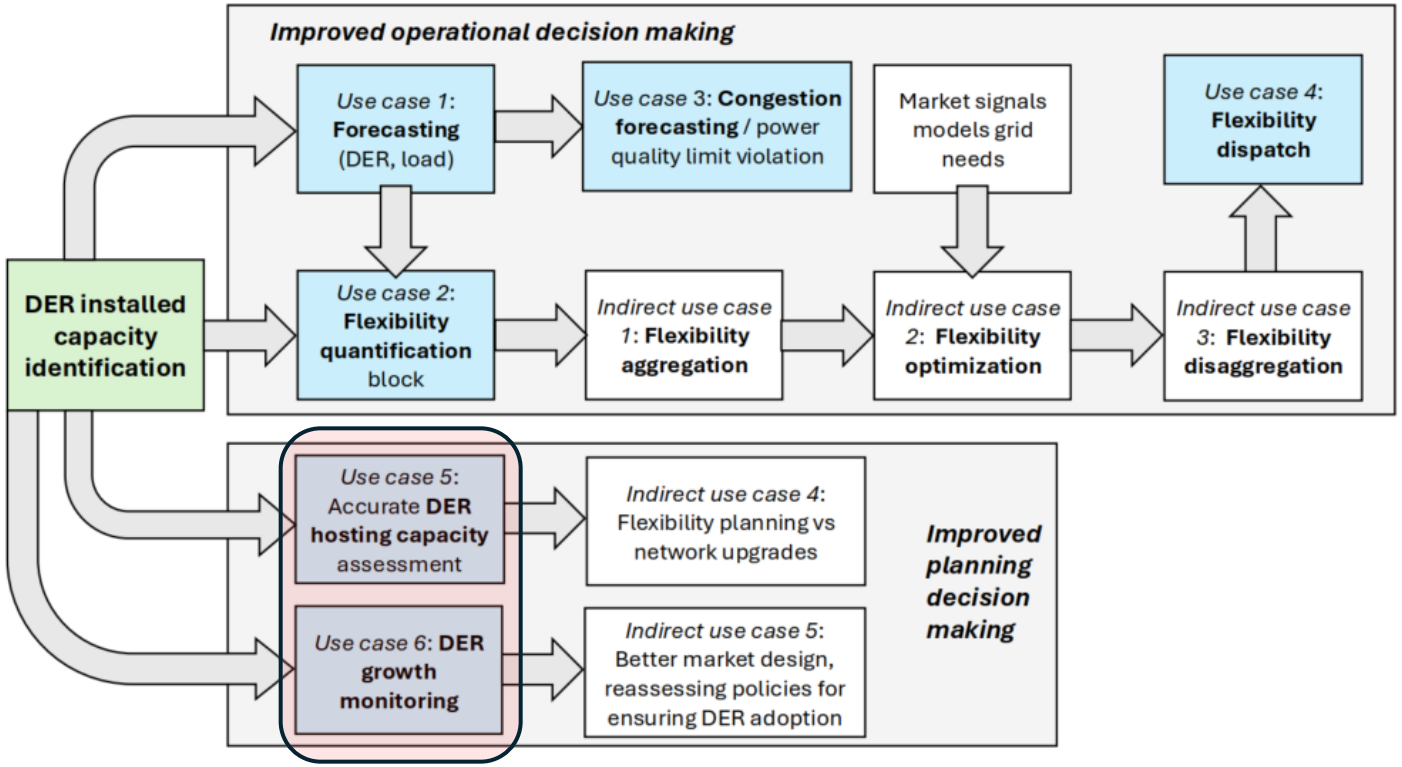
Use Cases 3–4: Flexibility Quantification & Flexibility Dispatch

- Installed DER capacity enables estimation of flexibility envelopes & feasibility regions
- Supports targeted flexibility activation for locational issues (congestion, voltage)



Use Cases 5–6: Hosting Capacity & DER Growth Monitoring

- Metadata-aware HCA replaces unrealistic greenfield assumptions; avoids over/under investment
- Periodic DER metadata estimation allows spatial tracking of DER adoption trends



Any questions?

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